

UNIVERSITY OF CANBERRA

HDR Research Proposal

Grid-less phantom breast imaging using thin aluminium sheets
for scatter reduction.

Diana Fifield email: u3033803@uni.canberra.edu.au

University of Canberra

July 2018

Degree: Honours!

Supervisor: Professor Rob Davidson 

Due: [May 2019]

Experiments
done

Explained in
detail.

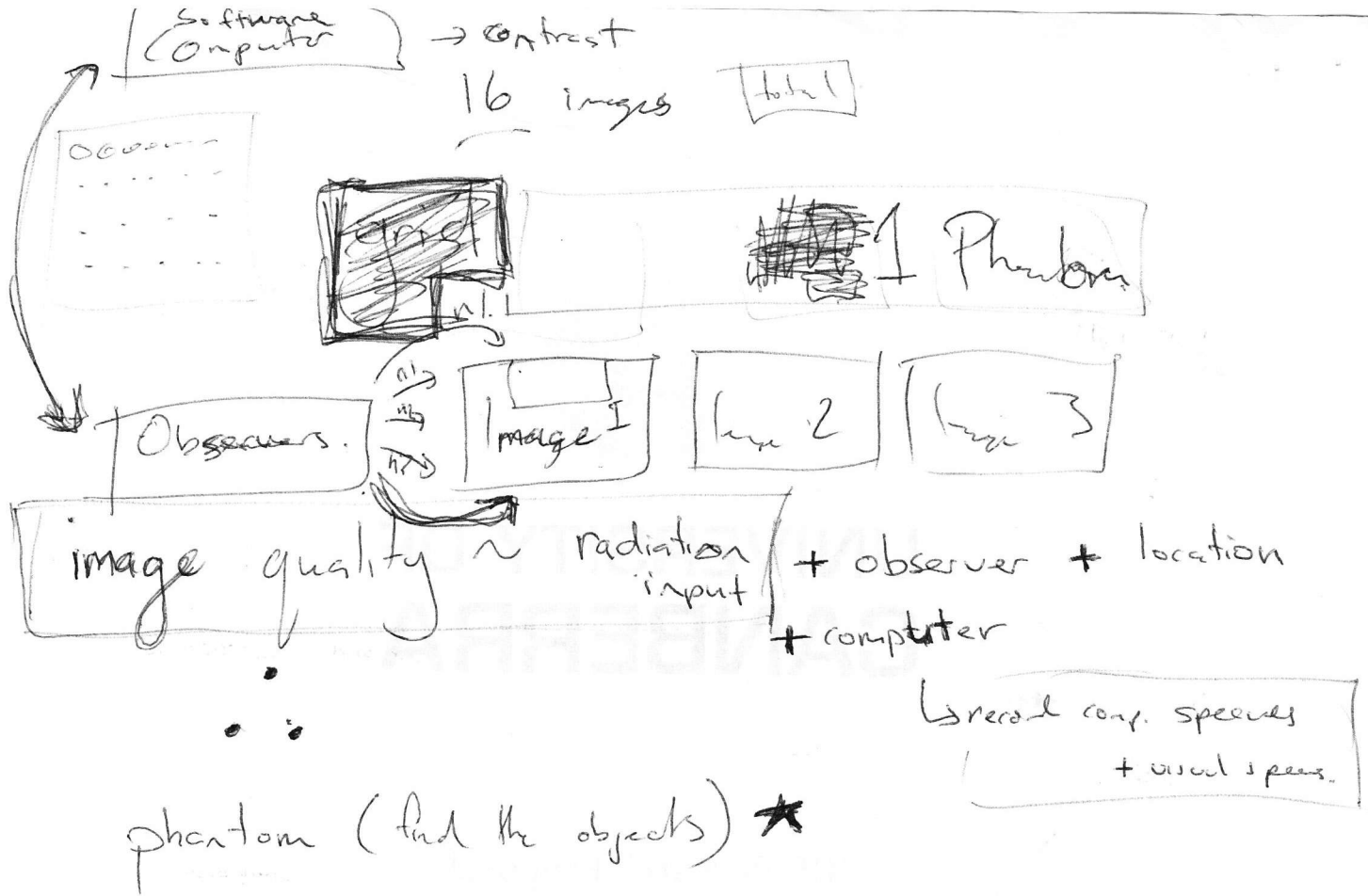


Image ← score ⇒ Outcome variable explained and defined.

Computer analysis

lit. review.

Observer analysis.

DATASET / Datastreet.

Pre data

Observer	Computer	Time start	Time finish	Radiation	Image	Dose
					...	-

Study aims and rationale

It is a priority in mammography to minimise ~~patient~~ ^{the} dose. This is balanced against the need

for optimal image quality. Scatter radiation is a common effect of the x-ray process, and

^{where we} grids are used in x-rays ^{procedures} to reduce the amount of ^{this harmful} scatter radiation. Scatter radiation ^{also damages grid by} reduces

image contrast in an x-ray image, masking detail and rendering images non-diagnostic. ~~The~~ ^{However,}

~~disadvantage of grid use, is~~ ^{grid use requires} an increased amount of radiation is required to create an x-ray

image, thereby increasing the dose to the patient. The aim of this study is to test the

effectiveness of using thin aluminium sheets as a source of attenuation and a replacement

for grids in mammography. It is speculated that the aluminium sheets will reduce scatter

radiation sufficiently to enable diagnostic quality images to be generated, whilst minimising

the amount of ionising radiation required. Expert observers will be used to visually grade

one group of the images for contrast and spatial resolution, and analytical software will be

used on the second group of images to assess for contrast resolution. A good outcome for

this study will be if a visually noticeable reduction in scatter radiation is detected.

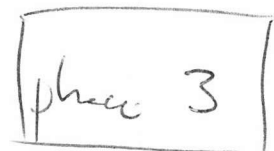
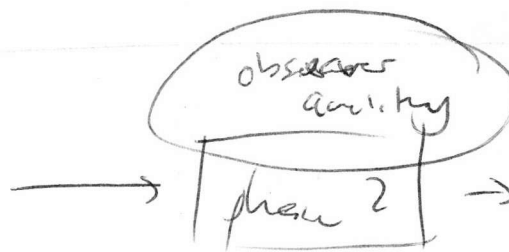
Research question

Can a diagnostically equivalent mammographic image be obtained by replacing the
mammography grid with thin aluminium sheets?

Method

Research Design

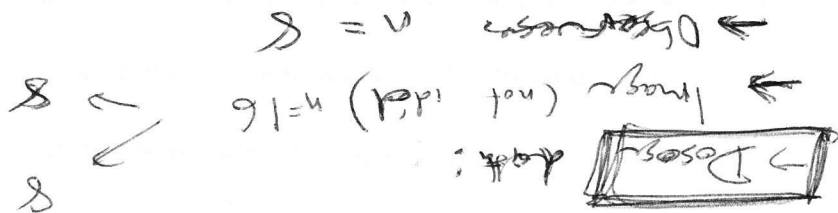
This study is a quantitative experimental study involving empirical data collection both in
generating x-ray images and subsequent visual grading analysis and ranking of these images
based on image quality. This is a three-phase project with two data collection phases and an
analysis phase.



LINK
with
methods

Methods:

Data collection



Analysis

Software

Replication in

Observations

Software

Sampling methods

Two sampling methods will be used to collect data in phases 1 and 2.

- Phase 1: data collection involves gathering data calculated by the mammographic imaging system at the time of each x-ray exposure.
- Phase 2: data collection is obtained subsequent to x-ray imaging and involves image analysis using either expert observers or inverse image quality figure (IQFInv) software.

Phase 1: Collection of raw data for analysis

Timeline: completed by end September 2018, with 1 day required to collect the data.

Cost: No upfront costs are expected for this part of the study; TCH is generously donating their equipment usage.

- Using a Hologic Selenia mammography system available at the Canberra Hospital (TCH), data for mAs and dose will be recorded based on the independent variables: kVp, filter type, contrast resolving phantom type and attenuating material (see table below).
- De-identified images will be recorded for each projection of both the ACR and Artinis phantom exposures for use in phase 2.

kVp	Filter	Phantom	Attenuating material	mAs	Various dosage readings	Image analysis technique
28	Rh	ACR	nil	AEC		observer
28	Rh	ACR	grid	AEC		observer
28	Rh	ACR	0.05mm Al	AEC		observer
28	Rh	ACR	0.1mm Al	AEC		observer

32	Ag	ACR	nil	AEC		observer
32	Ag	ACR	grid	AEC		observer
32	Ag	ACR	0.05mm Al	AEC		observer
32	Ag	ACR	0.1mm Al	AEC		observer
28	Rh	ARTINIS	nil	AEC		IQFinv
28	Rh	ARTINIS	grid	AEC		IQFinv
28	Rh	ARTINIS	0.05mm Al	AEC		IQFinv
28	Rh	ARTINIS	0.1mm Al	AEC		IQFinv
32	Ag	ARTINIS	nil	AEC		IQFinv
32	Ag	ARTINIS	grid	AEC		IQFinv
32	Ag	ARTINIS	0.05mm Al	AEC		IQFinv
32	Ag	ARTINIS	0.1mm Al	AEC		IQFinv

Phase 2: Collection of observer data

Timeline: completed by end November 2018

Cost: No upfront costs are expected for this part of the study; observers are generously }
donating their time.

Part I

- A total of 8 expert mammography observers; 4 from Department of Radiologic Sciences, Faculty of Allied Health Sciences, Kuwait University, Kuwait and 4 from Breast Screen Australia (Canberra), will be requested to complete visual grading analysis of each of the images relative to the randomised images based on visual contrast and spatial resolution.
- As experts, the observers will fully understand the requirements for quality diagnostic mammographic images. A structured observational checklist based on the

confounding effects

ACR scoring key for embedded artefacts within the phantom will be used. Artefacts consist of specks, fibres and masses in the ACR phantom which somewhat mimic cancerous lesions in appearance.

- For the purposes of consistency, observers will be requested to use the same monitor and/or computer throughout the observation process. All observers are not expected to use the same monitor. Observers will also be requested to view and report on all images, 8 in total, in a single sitting, with short breaks as needed. rgb
color
stache
- Expected time needed for grading is 30 - 90 minutes. → each image of total?
- A repeat reading will be requested two weeks after the first. (Ko et al., 2013). This will provide data for intra-observer analysis.
- The Artinis phantom uses decreasing size gold discs. This phantom will be used in combination with the Artinis CDMAM Analyser software, which is designed to identify low contrast objects in a radiographic image. The smallest sized object identified by the Analyser software will be recorded for each image.

Part II - Observer images only

- Observers will be asked to rank the images based on overall diagnostic quality.
- As part of the data collection, each observer will be asked to record:
 - i. time of day analysis took place (image fatigue)
 - ii. years of experience in mammographic image analysis.
- Observers will be permitted to adjust windowing levels to optimise image analysis, as would occur during mammographic image analysis.

Phase 3:

- i. statistical analysis of observer image analysis generated data (Phase 2 data).
- ii. data analysis of software analysed image data.

Timeline: completed end January 2019

Cost: Costs are unknown at this stage.

- Required analysis: intra-observer analysis

inter-observer analysis

comparative analysis of observer image scoring

comparative analysis of analytical software scoring

equating image quality to dose received.

Ethics

Ethics approval will be required for this study with regards the use of participant observers.

This is a very low risk study because participant involvement relates to the observer role only.

Risks & discomforts

There are no risks of discomforts. Participants may take short breaks as needed throughout the predicted observation period of up to 90 minutes.

Benefits

Recognition and possible personal development hours.

Confidentiality & Anonymity

It is expected that the participants will be named and acknowledged as observers on any publications that arise from this study. Prior permission will be obtained from individuals before any publications.

Compensation

No compensation is offered.

Contact

Australian contact: Diana Fifield

Kuwaiti contact: Khaled Alkalifah

Voluntary participation

All participants will be voluntary. Australian participants may be able to count the hours towards their annual personal development.

Escape clause

Participants are under no obligation to complete the entire observation and can leave at any time.

Ko, M.-S., Kim, H. H., Cha, J. H., Shin, H. J., Kim, J. H., & Kim, M. J. (2013). Dose Reduction in Automatic Optimization Parameter of Full Field Digital Mammography: Breast Phantom Study. *Journal of breast cancer*, 16(1), 90-96. doi: 10.4048/jbc.2013.16.1.90

Budget:

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START

FINISH

PAUSE



next